Vakil 28.4, 28.5 Reading Note.

28.4.1. Zariski's Connectedness Lemma. — *If a proper morphism $\pi\colon X \to Y$ of locally Noetherian schemes is $\mathcal{O}$-connected (see §25.1.7), then $\pi^{-1}(q)$ is connected for every $q \in Y$.*

i.e. $\mathcal{O}_Y \simeq \pi_* \mathcal{O}_X$

proven: all used for Formal Function Theorem.

Lemme

28.4.A. Easy Exercise (the completion of a local ring is a local ring). Suppose $(A, \mathfrak{m})$ is a Noetherian local ring. Show that the completion of A along $\mathfrak{m}$, $\hat{A} := \lim A/\mathfrak{m}^n$, is a local ring, with maximal ideal $\mathfrak{m}\hat{A}$. Hint: show that any element of $\hat{A}$ not in $\mathfrak{m}\hat{A}$ is invertible.

pf. • $\hat{A} \xrightarrow{\pi} A/\mathfrak{m}$. Claim: $\ker\pi = \mathfrak{m}\hat{A}$. $A \to \hat{A}$ by $a \mapsto (\bar{a}, \bar{a}, \bar{a}, \dots)$

reason: $\forall x \in \ker\pi$, denote $x = (\bar{a}_1, \bar{a}_2, \dots)$ with $a_i \in A$, $\bar{a}_i \in A/\mathfrak{m}^i$

by definition of $\lim$: $\hat{A} \to A/\mathfrak{m}^n$, $a_{n+1} - a_n \in \mathfrak{m}^n$.

Since A noetherian, write $\mathfrak{m} = (t_1, \dots, t_r)$

and $a_{n+1} - a_n = \sum_{i=1}^{r} b_{n,i} t_i$ with $b_{n,i} \in \mathfrak{m}^{n-1}$ $(n \geq 1)$

$\because x \in \ker\pi$, $\therefore a_1 \in \mathfrak{m}$, i.e. $a_1 - a_0 = \sum_{i=1}^{r} b_{1,i} t_i$ with $b_{1,i} \in A$.

consider $y_i = (\overline{b_{1,i}}, \overline{b_{1,i} + b_{2,i}}, \overline{b_{1,i} + b_{2,i} + b_{3,i}}, \dots) \in \hat{A}$ (easily verified)

then as element in $\hat{A}$:

$\sum_{i=1}^{r} y_i t_i = (\overline{a_1 - 0}, \overline{a_2 - 0}, \dots) = x$ i.e. $x \in \mathfrak{m}\hat{A}$.

$\therefore \mathfrak{m}\hat{A}$ is maximal ideal of $\hat{A}$.

• $(\hat{A} \setminus \mathfrak{m}\hat{A}) \subseteq \hat{A}^*$.

reason: $\forall x \in$ LHS $x = (\bar{a}_1, \bar{a}_2, \bar{a}_3, \dots)$ as above.

$a_1 \notin \mathfrak{m}$. $\therefore$ by $a_{n+1} - a_n \in \mathfrak{m}^n \subseteq \mathfrak{m}$. $a_n \notin \mathfrak{m}$ $\forall n \in \mathbb{N}^*$

$\because A$ local ring, $\therefore a_n \in A^*$ $\forall n \in \mathbb{N}^*$

consider $y = (\overline{a_1^{-1}}, \overline{a_2^{-1}}, \dots)$

well-definedness of y: $a_{n+1}^{-1} - a_n^{-1} = -a_{n+1}^{-1} a_n^{-1} \cdot (a_{n+1} - a_n) \in \mathfrak{m}^n$

$xy = 1_{\hat{A}}$. ∎

pf of Zariski's Connectedness Lemma:

prove by contradiction. Suppose $\exists q \in Y$ s.t. $\pi^{-1}(q)$ is not connected.

write $\pi^{-1}(q) = X_1 \sqcup X_2$ X_i open subscheme of $\pi^{-1}(q)$

After $-\otimes_Y \mathcal{O}_Y/\mathfrak{m}_q^n$ (i.e. nth order formal neighborhood of $\pi^{-1}(q)$), is still disconnected. denote $\pi^{-1}_{(n)}(q)$

Apply Formal Function Theorem:

28.3.2. Theorem on Formal Functions. — *Suppose $\pi\colon X \to Y$ is a proper morphism of locally Noetherian schemes, $\mathscr{F}$ is a coherent sheaf on X, and $q \in Y$. Then (28.3.1.2) is an isomorphism for all $i \geq 0$.*

$$(28.3.1.2) \qquad \left((R^i\pi_*\mathscr{F})_q\right)^{\wedge} \longrightarrow \lim_{n\to\infty} H^i(X_n, \mathscr{F}|_{X_n}).$$

(Take $i=0$, $\mathscr{F}=\mathcal{O}_X$ and using $\mathcal{O}$-connected condition)

$$\widehat{\mathcal{O}_{Y,q}} \xrightarrow{\cong} \lim_{n\to\infty} H^0(\pi^{-1}_{(n)}(q), \mathcal{O}_{\pi^{-1}_{(n)}(q)})$$

where $\pi^{-1}_{(n)}(q) = X_{1(n)} \sqcup X_{2(n)}$

LHS is local ring by previous Lemma.

RHS: for $m<n$,

$$\begin{array}{ccc} H^0(\pi^{-1}_{(n)}(q), \mathcal{O}) & \longrightarrow & H^0(\pi^{-1}_{(m)}(q), \mathcal{O}) \\ \| & & \| \\ \bigoplus_{i=1}^{2} H^0(X_{i(n)}, \mathcal{O}) & \longrightarrow & \bigoplus_{i=1}^{2} H^0(X_{i(m)}, \mathcal{O}) \\ (1,0) & \longmapsto & (1,0) \\ (0,1) & \longmapsto & (0,1) \end{array}$$

"$e_1 = ((1,0)(1,0)\ldots\ldots)$ $e_2 = ((0,1)(0,1)\ldots\ldots)$" $\in$ RHS.

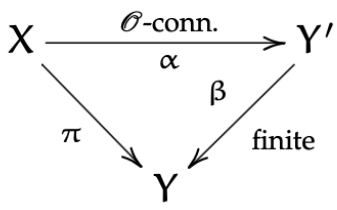
$e_1 \cdot e_2 = 0$. $e_1 + e_2 = 1$ By LHS $\cong$ RHS. see as elements in LHS

WMA $e_1 \in \mathfrak{m}$. $\Rightarrow e_2 \notin \mathfrak{m}$ $\Rightarrow e_2$ unit. $\Rightarrow e_1 = 0$. contradiction! ▨

(comment: easy application of formal function theorem). (Above condition $\mathcal{O}$-connected can be replaced by $\mathcal{O} \twoheadrightarrow \pi_*\mathcal{O}$ surj.)

28.4.3. Stein Factorization Theorem. — *Any proper morphism $\pi\colon X \to Y$ of locally Noetherian schemes can be factored into $\beta \circ \alpha$, where $\alpha\colon X \to Y'$ is proper and $\mathcal{O}$-connected (hence has connected fibers, by Zariski's Connectedness Lemma 28.4.1), and $\beta\colon Y' \to Y$ is a finite morphism.*

(28.4.3.1)



(proj) (proj) where "proper" used.

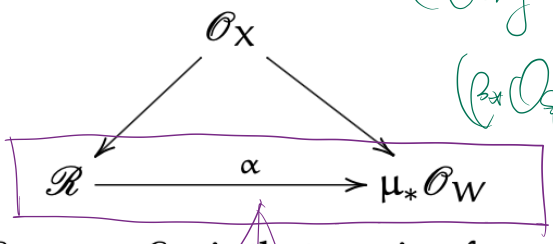
pf. Consider $Y' = \operatorname{Spec}_Y \pi_*\mathcal{O}_X$ $\pi_*\mathcal{O}_X$ is coherent $\mathcal{O}_Y$-module & $\mathcal{O}_Y$-algebra

hence $\operatorname{Spec}_Y \pi_*\mathcal{O}_X \xrightarrow{\beta} Y$ is finite.

recall

17.1.2. Second, it will satisfy a universal property. We could define the A-scheme Spec R by the fact that morphisms to Spec R (from an A-scheme W, over Spec A) correspond to maps of A-algebras $R \to \Gamma(W, \mathcal{O}_W)$ (this is our old friend Exercise 7.3.G). The universal property for $\beta\colon \mathit{Spec}\,\mathscr{R} \to X$ generalizes this. Given a morphism $\mu\colon W \to X$, the X-morphisms $W \to \mathit{Spec}\,\mathscr{R}$ are in functorial (in W) bijection with morphisms α of sheaves of algebras making

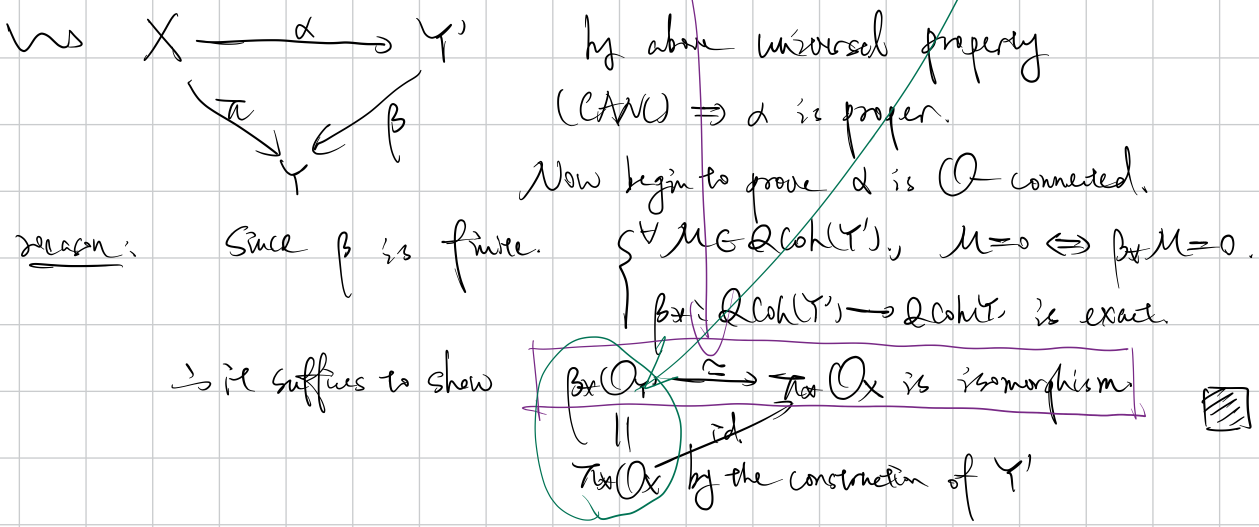
(17.1.2.1)



(Taking $\alpha = \mathrm{id}_{\mu_*\mathcal{O}_W}$)

($\beta_*\mathcal{O}_{\mathit{Spec}\,\mathscr{R}} = \mathscr{R}$)

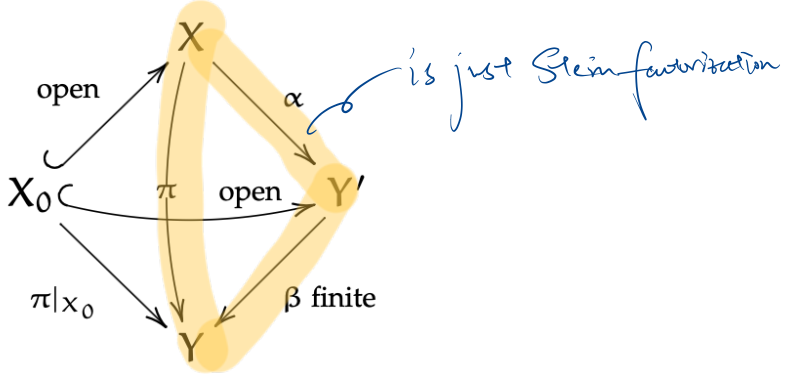
commute. Here the map $\mathcal{O}_X \to \mu_*\mathcal{O}_W$ is that coming from the map of ringed spaces, and the map $\mathcal{O}_X \to \mathscr{R}$ comes from the $\mathcal{O}_X$-algebra structure on $\mathscr{R}$. (For experts: it needn't be true that $\mu_*\mathcal{O}_W$ is quasicoherent, but that doesn't matter. Non-experts should completely ignore this parenthetical comment.)



28.5.1. Zariski's Main Theorem (Grothendieck's form). — *Suppose $\pi\colon X \to Y$ is a proper morphism of locally Noetherian schemes.*
(a) The set of points of X that are isolated in their fiber forms an open subset $X_0 \subset X$.
(b) The morphism $\pi|_{X_0}\colon X_0 \to Y$ factors into an open embedding $X_0 \hookrightarrow Y'$ followed by a finite morphism $Y' \to Y$.
(c) The morphism $\pi\colon X \to Y$ factors through Y'.

previously: $\{x \in X \mid (\Omega_{X/Y})_x = 0\}$

(28.5.1.1)



As with Stein factorization, the proof of the theorem yields a *specific* factorization. Indeed, $X \to Y' = \mathit{Spec}\,\pi_*\mathcal{O}_X \to Y$ *is* the Stein factorization of $\pi : X \to Y$, giving (c) and part of (b) basically immediately.

Before we start, we note that the fiber of a proper morphism over a point $q \in Y$ is a proper (hence finite type) scheme over the residue field $\kappa(q)$, and we say that a point is *isolated in its fiber* if it forms a component of the fiber of dimension 0. (Here component means both connected and irreducible component.)

i.e. $x \in X$ is isolated in its fiber $\Leftrightarrow$ $y = \pi(x)$, $X_y = C_1 \cup \cdots \cup C_r$ (irr. components) (by properness of $X_y \to \{y\}$) with $\{x\} = C_i$ for some $i \in [1, r]$.

($x \in C_i$ with $\dim C_i = 0 \Leftrightarrow \{x\} = C_i$)

(hence C_i closed w.r.t both connected component & irr. component)

pf Take Stein's factorization of $X \xrightarrow{\pi} Y$ as following:

$$\begin{array}{ccc} X & \xrightarrow{\alpha} & Y' \\ & \searrow^{\pi} \quad \swarrow_{\beta} & \\ & Y & \end{array}$$

Since β is finite. Normal Fact: $\forall x \in X$, x is isolated in fiber of $\pi \Leftrightarrow x$ is isolated in fiber of α

By Zariski's connectedness Lemma. $\forall x \in X$. x is isolated in fiber of $\alpha \Leftrightarrow \alpha^{-1}(\alpha(x)) = \{x\}$

Claim: $\alpha^{-1}(\alpha(x)) = \{x\} \Leftrightarrow \Omega_{X/Y} \otimes_{\mathcal{O}_{X,x}} \kappa(x) = i_x^* \Omega_\alpha|_x = 0$.

reason: "$\Leftarrow$": If $i_x^* \Omega_\alpha|_x = 0$. $\alpha(x) = y'$. η be the generic pt. of the irreducible component of $X_{y'}$ containing x.

$$0 = i_x^* \Omega_\alpha|_x = \Omega_{X_{y'}/\kappa(y')} \otimes_{\mathcal{O}_{X_{y'},x}} \kappa(x) \qquad X_{y'} \to \{y'\} \text{ proper.}$$

By Nakayama's Lemma. $\Omega_{X_{y'}/\kappa(y'),x} = 0$ as $\mathcal{O}_{X_{y'},x}$-module.

$\mathcal{O}_{X_{y'},x} \to \kappa(\eta)$ is normal. $\therefore \Omega_{X_{y'}/\kappa(y')} \otimes_{\mathcal{O}_{X_{y'},\eta}} \kappa(\eta) = 0$.

By $\Omega_{X_{y'}/\kappa(y')} \otimes_{\mathcal{O}_{X_{y'},\eta}} \kappa(\eta) \to \Omega_{\kappa(\eta)/\kappa(y')} \to \Omega_{\kappa(\eta)/X_{y'}} \to 0$

$\therefore \Omega_{\kappa(\eta)/\kappa(y')} = 0$

$\Omega_{\kappa(\eta)/X_{y'}} = 0$ i.e. $\begin{array}{ccc} \kappa(\eta) & \to & \kappa(\eta) \\ \downarrow & \square & \downarrow \\ \kappa(y') & \to & X_{y'} \end{array}$

Recall: L/k fin gen. field extn. $\Omega_{L/k} = 0 \Leftrightarrow L/k$ finite separable.

$\therefore \kappa(\eta)/\kappa(y')$ is fin. separable extension.

$\therefore \eta$ is a closed pt. in $X_{y'}$ over $\kappa(y')$ and $x = \eta$. i.e. x is isolated in fiber of α

5.3.F. EXERCISE. Show that a point of a locally finite type k-scheme is a closed point if and only if the residue field of the stalk of the structure sheaf at that point is a finite extension of k. Show that on a locally finite type k-scheme, the closed points are dense. Hint: §3.6.9. (Warning: closed points need not be dense even on quite reasonable schemes, see Exercise 3.6.J(b).)

"$\Rightarrow$" If $\alpha(x) = y'$, $\alpha^{-1}(y') = \{x\}$. Claim: $\mathcal{O}_{Y',y'} \xrightarrow{\alpha^\sharp} \mathcal{O}_{X,x}$ is isomorphism.

reason: $\mathcal{O}_{Y'} \xrightarrow{\cong} \alpha_* \mathcal{O}_X$

$$\mathcal{O}_{Y',y'} = \varinjlim_{V \ni y} \Gamma(V, \mathcal{O}_{Y'}) \cong \varinjlim_{V \ni y} \Gamma(\alpha^{-1}(V), \mathcal{O}_X)$$

To show $\varinjlim_{V\supset y} \Gamma(\alpha^{-1}(V), \mathcal{O}_X) = \varinjlim_{U\supset x} \Gamma(U, \mathcal{O}_X) = \mathcal{O}_{X,x}$.

It suffices to show that $\forall U\ni x.\ \exists V\ni y$ s.t. $\alpha^{-1}(V)\subseteq U$.

$X\setminus U$ closed. $\therefore$ $\alpha(X\setminus U)$ closed in Y' since α is proper

consider $V = Y'\setminus \alpha(X\setminus U)$. then by definition $\alpha^{-1}(V)\subseteq U$.

Suppose $y'\in \alpha(X\setminus U)$. then contradicts with $\alpha^{-1}(y') = \{x'\}$. $\therefore$ $V\ni y'$. ✓

Now we have $\mathcal{O}_{Y,y} \xrightarrow{\alpha} \mathcal{O}_{X,x}$ isomorphism.

$\therefore$ (c) Let U be an open subset of X; then $\Omega^1_{X/Y}|_U \simeq \Omega^1_{U/Y}$. For any $x \in X$, we have $(\Omega^1_{X/Y})_x \simeq \Omega^1_{\mathcal{O}_{X,x}/\mathcal{O}_{Y,f(x)}}$. (prop 6.1.24 Liu Qing)

$(\Omega_{X/Y'})_x = \Omega_{\mathcal{O}_{X,x}/\mathcal{O}_{Y',y'}} = 0$. $\therefore$ $\Omega_\alpha|_x = 0$.

$\therefore$ $X_0 = \{x\in X \mid (\Omega_{X/Y'})|_x = 0\}$ is open in X. (a) ▨

For (b).

① $Y'\setminus \alpha(X\setminus X_0)$ is open in Y' by α proper

$X_0 \xrightarrow{\tilde{\alpha}} Y'\setminus\alpha(X\setminus X_0)$ well-defined as map between sets by

② To show $\tilde{\alpha}$ is homeomorphism. $\tilde{\alpha}^{-1}(\text{open}) = \text{open}$ inherited from α

It suffices to show $\alpha(\text{open}) = \text{open}$.

$\forall U\subseteq X_0$ open. $\alpha(X\setminus U)$ closed in Y'. Claim: $\alpha(U) = Y'\setminus\alpha(X\setminus U)$, hence open.

reason: **25.1.H.** EASY EXERCISE. Show that proper $\mathcal{O}$-connected morphisms of locally Noetherian schemes are surjective.

$\alpha(U) \subseteq Y'\setminus\alpha(X\setminus U) \subseteq \alpha(U)$

reason as above — by 25.1.H

③ $(X_0, \mathcal{O}_{X_0}) \cong (\underbrace{Y'\setminus\alpha(X\setminus X_0)}_{\| Y_0}, \mathcal{O}_{Y_0})$.

$\because \mathcal{O}_{Y_0}(\alpha(U)) = \mathcal{O}_{Y'}(\alpha(U)) \cong \alpha_*\mathcal{O}_X(\alpha(U)) = \mathcal{O}_X(\alpha^{-1}\alpha(U)) = \mathcal{O}_X(U)$ compatible with $U\subseteq U'$, $\forall U\in \mathrm{Top}(X_0)$ ▨

(c) from our construction ▨

pf of 25.1.H: Suppose $Y\setminus\pi(X) \neq 0$. π proper $\therefore$ $Y\setminus\pi X$ open

$\mathrm{Spec}A \subseteq Y\setminus\pi X$ affine non-trivial

$A = \Gamma(\mathrm{Spec}A, \mathcal{O}_Y) \cong \Gamma(\mathrm{Spec}A, \pi_*\mathcal{O}_X) = \Gamma(\pi^{-1}(\mathrm{Spec}A), \mathcal{O}_X) = 0$.

contradiction! ▨

28.4.B. EXERCISE. Suppose $\pi\colon X \to Y$ is a proper birational morphism of irreducible varieties. Suppose further that Y is normal. Show that π is $\mathcal{O}$-connected. (This applies, for example, to blow-ups of smooth varieties under smooth centers.) Hence, by Zariski's Connectedness Lemma 28.4.1, π has connected fibers. This is called **Zariski's Main Theorem** (the version for birational morphisms). Hint: Exercise 10.7.P.

pf $\pi_*\mathcal{O}_X$ is coherent $\mathcal{O}_Y$-module & $\mathcal{O}_Y$-algebra.

$\therefore \forall U \subseteq Y$ affine $\Gamma(U, \pi_*\mathcal{O}_X)$ is finite $\Gamma(U,\mathcal{O}_Y)$-algebra

$$\begin{array}{ccc} \Gamma(U,\mathcal{O}_Y) & \hookrightarrow & \Gamma(U,\pi_*\mathcal{O}_X) \\ \downarrow & & \downarrow \\ K(Y) & \xrightarrow{\cong} & K(X) \end{array}$$

since π is birational, & X, Y irreducible var.

By normality of Y. $\Gamma(U,\mathcal{O}_Y) \to \Gamma(U,\pi_*\mathcal{O}_X)$ can only be isomorphism.

$\therefore \mathcal{O}_Y \xrightarrow{\cong} \pi_*\mathcal{O}_X$, i.e. π is $\mathcal{O}$-connected. ▨

Can be removed but correct by Y, Y' normal $\Rightarrow \mathcal{O}$-connected.

28.4.C. EXERCISE. Suppose $\pi\colon X \to Y$ and $\pi'\colon X \to Y'$ are two proper birational morphisms of integral locally Noetherian schemes, each contracting the same *connected* set of X to a point (and an isomorphism elsewhere), and suppose Y and Y' are normal. Show that π and π' are the same (or more precisely, that there is an isomorphism $i\colon Y \xrightarrow{\sim} Y'$ such that $\pi' = i \circ \pi$). Informal translation: If $X \to Y$ has connected fibers, then Y is determined by just knowing the locus contracted on X. Hint: Identify Y and Y' as topological spaces, and then identify π and π' as maps of topological spaces. Use Stein factorization to recover the structure sheaf as $\pi_*\mathcal{O}_X$.

pf Define $i: Y \to Y'$ by $y \mapsto \pi'(\pi^{-1}y)$ set-theoretically

well-defined by our condition

Similarly we have $i^{-1}: Y' \to Y$: $y' \mapsto \pi(\pi'^{-1}y')$

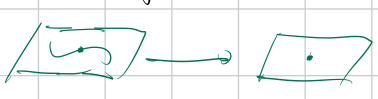
and i, i^{-1} form bijection set-theoretically

- i, i^{-1} form isomorphism in Zariski topology., mainly because π, π' are closed (from proper) and surjective (from proper & birational), details omitted.
- i, i^{-1} form isomorphism of schemes. (using Y, Y' normal here)

By previous Exercise. π, π' are $\mathcal{O}$-connected.

$\therefore \mathcal{O}_Y(U) \cong \pi_*\mathcal{O}_X(U) = \mathcal{O}_X(\pi^{-1}U) = \mathcal{O}_X(\pi'^{-1}(iU)) = \pi'_*\mathcal{O}_X(iU) \cong \mathcal{O}_{Y'}(iU)$

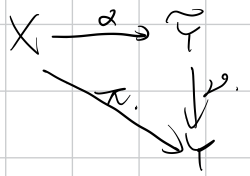
and they are compatible with $U \subseteq V$ in $\mathrm{Top}(Y)$ ▨

meaning of our condition:  isomorphism on $Y \setminus \{\text{one closed pt}\}$

Small addition:

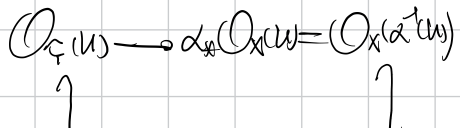
prop $\pi: X \to Y$ proper surj. morphism of irr. varieties over k
If π is $\mathcal{O}$-connected and X is normal, then Y is normal.

pf



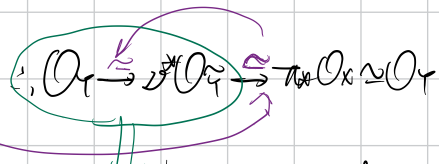
$\tilde{Y}$ is normalization of Y. ν is finite.
α comes from the universal property of normalization.
α is proper ($\because$ (CANC) ν, π proper)

surj. (suppose not surj. $\alpha(X) \subsetneq \tilde{Y}$ $\tilde{Y}$ irreducible.
$\therefore \dim \alpha(X) < \dim \tilde{Y}$. $\therefore \dim Y = \dim f(X) = \dim \nu(\alpha(X))$
$\leq \dim \alpha(X) < \dim \tilde{Y} = \dim Y$
contradiction!)



$\Rightarrow \mathcal{O}_{\tilde{Y}} \to \alpha_* \mathcal{O}_X$ inj.

$\Rightarrow \nu_* \mathcal{O}_{\tilde{Y}} \to \pi_* \mathcal{O}_X$ inj.



$\therefore Y \cong \tilde{Y}$, Y is normal ∎

prop $\pi: X \to Y$ proper surj. morphism of normal irr. varieties.
then π is $\mathcal{O}$-connected $\Leftrightarrow$ $K(Y)$ is alg. closed in $K(X)$

pf "$\Rightarrow$" if $\mathcal{O}_Y \cong \pi_* \mathcal{O}_X$ if $\alpha \in K(X)$ is algebraic over $K(Y)$

$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_1 \alpha + a_0 = 0$. $a_i \in K(Y)$.

Choose $U \subseteq Y$ s.t. $a_i \in \Gamma(U, \mathcal{O}_Y)$ i.e. regular on U.

$\forall x \in f^{-1}(U)$, then $\mathcal{O}_{X,x}$ is integral closed & $a_{n-1}, \ldots, a_0 \in \mathcal{O}_{X,x}$.

used X normal

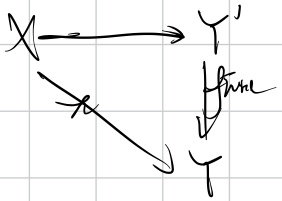
$\therefore \alpha \in \mathcal{O}_{X,x}$ $\therefore \alpha \in \bigcap_{x \in f^{-1}(U)} \mathcal{O}_{X,x} = \Gamma(f^{-1}(U), \mathcal{O}_X) = \Gamma(U, \mathcal{O}_Y) \subseteq K(Y)$

(这个等号是Hartogs (因为 X 是normal 所以 $\dim X \geq 2$) 这个等号是integral closed的性质)

by $\mathcal{O}$-connected

i.e. $K(Y)$ algebraically closed in $K(X)$

"$\Leftarrow$". if $K(Y)$ algebraically closed in $K(X)$



Stein factorization of $X \to Y$.

$K(Y')/K(Y)$ fin. extn. $\therefore$ can only be $K(Y') = K(Y)$

analytic affine locally $\Rightarrow Y' \cong Y$ normal

used Y normal

(more generally X normal have "$\Rightarrow$"
Y normal have "$\Leftarrow$")

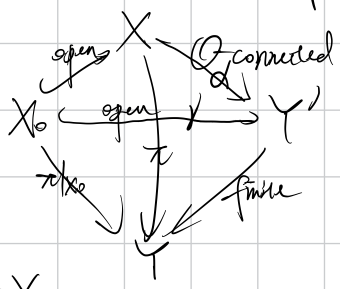
28.5.2. Theorem. — *Suppose $\pi\colon X \to Y$ is a morphism of locally Noetherian schemes. The following are equivalent.*

(a) *π is finite.*
(b) *π is affine and proper.*
(c) *π is proper and quasifinite.*

Preview: where we'll use ZMT? (c)⇒(a).

Pf of (c)⇒(a): If $\pi: X \to Y$ is proper and quasi-finite.

By ZMT



Observation: $X_0 = X$,

reason: By our proof in ZMT, $X_0 = \{x \in X \mid (\Omega_{X/Y})_x = 0\}$ (proof: finite type)
$= \{x \in X \mid \vec{\alpha}(\alpha x) = x\}$

$\forall x \in X$, since π quasi-finite $\Rightarrow \vec{\alpha}(\alpha x)$ is finite discrete

Since $\vec{\alpha}(\alpha x)$ is connected, this can only be $\vec{\alpha}(\alpha x) = x$. i.e. $X_0 = X$

Now $X \xrightarrow[\text{open}]{\text{connected}} Y'$ (CAN) ⇒ α is proper.

$\pi \searrow \quad \swarrow$ finite
Y

$\therefore \alpha X \hookrightarrow Y'$ open immersion w/ αX closed in Y'

Lem $\alpha X \hookrightarrow Y'$ above is actually closed immersion.

(reason: $Y' = \alpha X \sqcup (Y' \setminus \alpha X)$ form an open covering of Y'.
closed immersion is locally on the target. (simple recall: if closed imm. locally on the target ⇒ aff. loc. on the target ⇒ aff. ✓ by aff. ⇒ $O_Y \to f_* O_X$ surj. ⇒ closed immersion ⇒ closed imm.)
(i.e. $Z \hookrightarrow Y'$ closed immersion ⇔ $Z \cap \alpha X \hookrightarrow \alpha X$ are closed
$Z \times_{Y'} \alpha X$ $Z \cap (Y' \setminus \alpha X) \hookrightarrow Y' \setminus \alpha X$ immersions

⇒ $X \xrightarrow[\text{isom}]{\cong} \alpha X \hookrightarrow Y' \xrightarrow{\text{finite}} Y$ ∎
finite from closed imm.
finite by composition.

Appendix

Lemma 0389. *Let k be a field. Let X be a scheme over k. The following are equivalent*

(1) *X is geometrically connected,*

(2) *for every finite separable field extension k'/k the scheme $X_{k'}$ is connected.*

- Additional remark on Zariski's Connectedness Lemma:
  we can moreover prove that if $\mathcal{O}_Y \to \pi_*\mathcal{O}_X$ surjective, then every geometric fibre are connected.

Fact: $\forall y \in Y$ k/k sep. separable we can find $U \xrightarrow{g} Y$ étale & $u \in U$ s.t. $g(u)=y$ and $\kappa(u)=L$.

$$\begin{array}{ccc} V & \xrightarrow{\tilde{\pi}} & U \\ \tilde{g} \downarrow & \square & \downarrow g \text{ flat} \\ X & \xrightarrow[\text{proper}]{\pi} & Y \end{array}$$

by flat base change.

$$g^*(\pi_*\mathcal{O}_X) \xrightarrow{\simeq} \tilde{\pi}_*(\tilde{g}^*\mathcal{O}_X) = \tilde{\pi}_*\mathcal{O}_V$$

$$\therefore \mathcal{O}_U = g^*\mathcal{O}_Y \to g^*\pi_*\mathcal{O}_X \xrightarrow{\simeq} \tilde{\pi}_*\mathcal{O}_V$$

$\tilde{\pi}$ proper, V, U locally Noetherian.

Hence reduced to prove "fibres are connected" case, else see next page ▨ almost same as ⓪ connected case.

(Reference: Stacks project)

Lemma 02LF. *Let S be a scheme. Let $s \in S$. Let $k/\kappa(s)$ be a finite separable field extension. Then there exists an étale neighbourhood $(U, u) \to (S, s)$ such that the field extension $\kappa(u)/\kappa(s)$ is isomorphic to $k/\kappa(s)$.*

Proof. We may assume S is affine. In this case the lemma follows from Algebra, Lemma 00UD. □

(The reason I want to generalize is because of reading Kawamata's book and finding that the proof of Connectedness Lemma needs generalized conclusion, of $\bar{k}=\bar{k}$-var / k-var with $\mathrm{char}\, k=0$)

- ~~Fact~~: $h: X \to Y$ proper surj. Y normal. h connected fibers, then h connected geom. fibers.

pf. $X \xrightarrow{\alpha} Y' \xrightarrow{\beta} Y$ Stein factorization. β finite.

$\Omega_{Y'/Y} \cdot \eta_{Y'} = \Omega_{K(Y')/K(Y)} = 0$ (by $\mathrm{char}\, k=0$ $K(Y')/K(Y)$ fin. sep.)

$$\begin{array}{ccc} \{\eta_{Y'}\} & \to & Y' \\ \downarrow & \square & \downarrow \\ \{\eta_Y\} & \to & Y \end{array}$$

β fin surj. generically flat.

$\therefore \exists V \subseteq Y$ dense open s.t. $\beta^{-1}(V) \to V$ fin. étale.

In particular $\Omega_{\beta^{-1}(V)/V} = 0$.

$\therefore \forall y \in V$. $\beta^{-1}(y)$ are of the form $\coprod \mathrm{Spec}\, \Omega_i$ with Ω_i/k sep. fin. exten.

& $\sum [\Omega_i : k(y)] = \deg y = \dim_{k(y)} (\beta_* \mathcal{O}_Y \otimes k(y))$

deg is upper semicontinuous $\Rightarrow$ WMA $\deg y \equiv [K(Y):K(X)]$, $\forall y \in V$

When $k = \bar{k}$, consider y is k-pt. then it can only be $\beta^{-1}(y) = \underbrace{\mathrm{Spec}\, k \amalg \cdots \amalg \mathrm{Spec}\, k}_{[K(Y):K(X)]}$

disconnected, but $\alpha^{-1}(y)$ connected. contradiction

When k in general consider $\beta^{-1}(y) \times_{k(y)} \overline{k(y)}$ relative diff / $\overline{k(y)}$ still 0.

$\Rightarrow$ as above of the form $\underbrace{\mathrm{Spec}\, \overline{k(y)} \amalg \cdots \amalg \mathrm{Spec}\, \overline{k(y)}}_{[K(Y):K(X)]}$ disconnected.

contradiction!

(Rmk: if char $k = p$. counterexample: $\mathbb{A}^1_k \xrightarrow{h} \mathbb{A}^1_k$, $f \mapsto f^p$, fin. surj.

fiber single pt. But rank $h_* \mathcal{O}_X = p$. cant be $h_* \mathcal{O}_X \cong \mathcal{O}_Y$.)

(inspired by Jiang's HW Exercise 20).

- surjectivity of $(\mathcal{O}_{Y,y})^\wedge \to (\alpha_* \mathcal{O}_X)_y)^\wedge$ when $\mathcal{O}_Y \to \alpha_* \mathcal{O}_X$ surj.

Suppose that R is a ring and I is an ideal. We define the *completion of R with respect to I* to be the limit

$$R^\wedge = \lim_n R/I^n.$$

Lemma 0315. *Let R be a ring. Let $I \subset R$ be an ideal. Let $\varphi : M \to N$ be a map of R-modules.*

(1) *If $M/IM \to N/IN$ is surjective, then $M^\wedge \to N^\wedge$ is surjective.*
(2) *If $M \to N$ is surjective, then $M^\wedge \to N^\wedge$ is surjective.*
(3) *If $0 \to K \to M \to N \to 0$ is a short exact sequence of R-modules and N is flat, then $0 \to K^\wedge \to M^\wedge \to N^\wedge \to 0$ is a short exact sequence.*
(4) *The map $M \otimes_R R^\wedge \to M^\wedge$ is surjective for any finite R-module M.*

$(\mathcal{O}_{Y,y})^\wedge \to (\alpha_* \mathcal{O}_X)_y)^\wedge \cong \varprojlim_{n \in \mathbb{N}} H^0(n)$

$\tilde{e}_1 \mapsto e_1$, $\tilde{e}_2 \mapsto e_2$ existence of $\tilde{e}_i$ by cmj.

$\tilde{e}_1 + \tilde{e}_2 = 1$, $\tilde{e}_1 \tilde{e}_2 \in \ker \varphi$, if $\tilde{e}_1$ unit, then $\tilde{e}_2 \in \ker \varphi$, i.e. $e_2 = 0$ contradiction

$\Rightarrow \tilde{e}_1, \tilde{e}_2 \in m_1 \Rightarrow 1 \in m$ contradiction.

Exercise 20. Let $h : X \to Y$ and $h' : X \to Y'$ be projective morphisms with connected fibers between normal varieties. Assume that a curve C on X is contracted by h (i.e., $h(C)$ is a point) if and only if C is contracted by h'.

Show that there exists an isomorphism $\tau : Y \to Y'$ commuting with h, h'.

Sketch: $X \to Y \times Y'$ proper. $X \xrightarrow{(h,h')} Y \times Y'$

I := closed subvar. of $Y \times Y'$, integral.

$I \to Y$ finite

(Pf: For $y \in Y$ closed, $\dim h'(h^{-1}(y)) = 0$.
otherwise choose η a gen. pt. of an irr. comp. of $h'(h^{-1}(y))$.
$\xi \in h'^{-1}(\eta)$. $\overline{h(\xi)}$ is a curve. $h(\xi) = y$.
But $h(\xi) = y$. contradiction!)

$X \to I \xrightarrow{\text{finite}} Y$ proper surj. $\Rightarrow$ By above $I \to Y$ birational.
comm. fib. $\Rightarrow I \cong Y$. Similarly $I \cong Y'$. $\therefore Y \cong Y'$ ∎

(本质上还是这个用 Zariski's Main Theorem).
(X 为 normal).